

Wound Domain Ontology Specification

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Knowledge Representation & Reasoning – Assignment 2

Overview

A compact OWL ontology modeling **wound types**, **causes**, **symptoms**, **complications**, **treatments**, and **affected body parts**. Built in Protégé; supports reasoning over wound classification, risk, and care pathways.

□ **Body_Part_Wounded**

- Blood_Vessel
- Bone
- Ligament
- Muscle
- Nerve
- Skin
- Subcutaneous_Tissue
- Tendon

□ **Complication**

- Infection
 - Cellulitis (disjoint with Gas_Gangrene, Tetanus)
 - Gas_Gangrene (disjoint with Pyogenic_Infection, Tetanus)
 - Necrotizing_Fasciitis
 - Pyogenic_Infection (disjoint with Tetanus)
 - Sepsis
 - Tetanus
- Structural_Damage
 - Fracture
 - Nerve_Damage
 - Organ_Damage
- Systemic
 - Hemorrhage
 - Shock

□ **Symptom**

- Local_Sign

- Bleeding
 - Odor
 - Pain
 - Pus
 - Redness
 - Swelling
 - Tenderness
 - Warmth
- Systemic_Sign (disjoint with Local_Sign)
 - Fever
 - Muscle_Spasm
 - Shock_Symptom
 - Dizziness
 - Pale_Skin
 - Rapid_Heartbeat
 - Weakness

□ Treatment

- First_Aid
 - Bandaging
 - Cleaning
 - Direct_Pressure
 - Elevation
- Medical_Procedure
 - Amputation
 - Debridment
 - Drainage
 - Reattachment
- Medication
 - Antibiotic
 - Antiseptic
 - Pain_Relief
 - Tetanus_Vaccination
- Wound_Closure
 - Adhesive_Strips
 - Packing
 - Skin_Glue
 - Staples
 - Sutures

□ Wound_Cause

- Causal_Object
 - Blunt_Object
 - Projectile

- Rusty_Object
 - Scalpel
 - Sharp/Pointy_Object (disjoint with Blow)
- Injury_Mechanism
 - Bite
 - Blow
 - Deceleration
 - Friction
 - Twisting

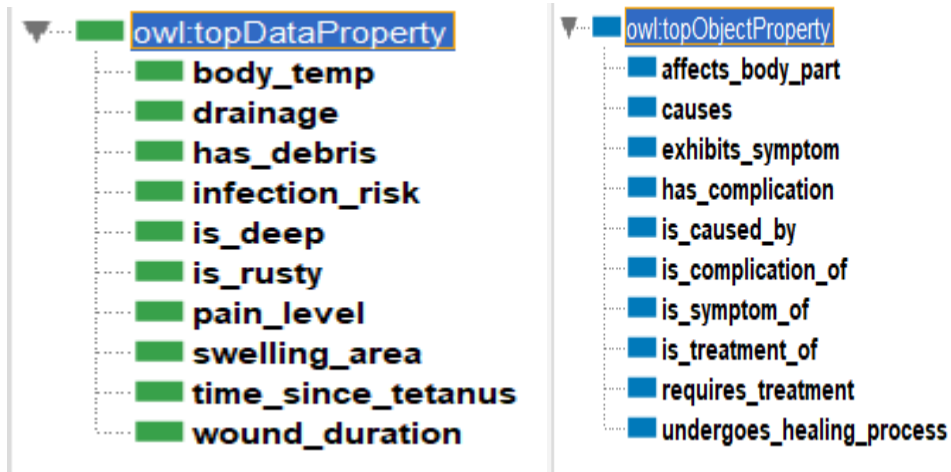
□ Wound_Healing_Process

- Clotting
- Contraction (disjoint with Epithelialization)
- Epithelialization (disjoint with Granulation)
- Granulation
- Scab_Formation
- Scar_Formation

□ Wound_Type

- Closed_Wound (disjoint with Open_Wound)
 - Constusion
 - Crush_Injury
 - Hematoma
 - Internal_Injury
- Open_Wound
 - Abrasion
 - Avulsion
 - Bite_Wound (disjoint with Closed_Wound, Surgical_Wound)
 - Gunshot/Shrapnel/Blast
 - Laceration
 - Puncture
 - Surgical_Wound (equivalent to Open_Wound and is_caused_by some Scalpel)
 - Class_I_(Clean)
 - Class_II_(Clean-Contaminated)
 - Class_III_(Contaminated)
 - Class_IV_(Dirty_Contaminated)

Data and object properties



Key Axioms

Axiom	Meaning
$\text{causes} \equiv \text{is_caused_by}^{-}$	Bidirectional: $\text{cause} \rightarrow \text{wound} \leftrightarrow \text{wound} \rightarrow \text{cause}$
$\text{exhibits_symptom} \equiv \text{is_symptom_of}^{-}$	Wound/complication shows symptom $\leftrightarrow$ symptom belongs to wound/complication
$\text{requires_treatment} \equiv \text{is_treatment_of}^{-}$	Wound needs treatment $\leftrightarrow$ treatment applies to wound
$\text{has_complication} \equiv \text{is_complication_of}^{-}$	Wound has complication $\leftrightarrow$ complication arises from wound
Domain/Range	Every Wound_Type must have: $\text{is_caused_by} \rightarrow \text{Wound_Cause}$, $\text{requires_treatment} \rightarrow \text{Treatment}$, $\text{affects_body_part} \rightarrow \text{Body_Part_Wounded}$
Functional Data Props	body_temp, is_deep, swelling_area, time_since_tetanus, wound_duration $\rightarrow$ one value only
$\text{Surgical_Wound} \equiv \text{Open_Wound} \sqcap \exists \text{is_caused_by.Scalpel}$	Surgical wounds are open and caused by scalpel
$\text{Open_Wound} \perp \text{Closed_Wound}$	No overlap between open and closed wounds
Disjoints	Surgical classes (I–IV), infection types, and local/systemic signs are mutually exclusive

Sample DL Queries

#	Purpose	DL Query	What it returns / proves
1	All wounds that are HIGH-RISK for infection	Wound_Type and (has_complication some Infection or infection_risk value true or is_caused_by some Rusty_Object or time_since_tetanus value "60"^^xsd:integer)	puncture_rusty_nail, gunshot_surgical_repair → shows risk inference from cause + data
2	Wounds that need ANTIBIOTICS (direct + inferred)	Wound_Type and (requires_treatment some Antibiotic or has_complication some Infection or exhibits_symptom some Pus or exhibits_symptom some Odor)	gunshot_surgical_repair, puncture_rusty_nail (even if not directly asserted, reasoner infers it)
3	Critical wounds needing SURGERY / MEDICAL PROCEDURE	Wound_Type and (requires_treatment some Medical_Procedure or has_complication some Structural_Damage or affects_body_part some (Muscle or Bone or Tendon or Nerve))	explosion_avulsion, gunshot_surgical_repair → deep tissue involvement triggers surgery
4	Wounds requiring TETANUS VACCINATION	Wound_Type and (is_caused_by some Rusty_Object or time_since_tetanus value "60"^^xsd:integer or has_complication value tetanus)	puncture_rusty_nail (catches both rusty cause and overdue shot)
5	Full emergency treatment bundle (what to do right now)	Treatment and is_treatment_of some (Wound_Type and (has_complication some Infection or infection_risk value true or wound_duration value "2"^^xsd:integer))	antibiotic_therapy, drainage_procedure, tetanus_vaccine, debridement → complete care plan

File: wound-ontology.owl

Reasoner: HermiT (consistent)

Purpose: Classify wounds → infer risks → recommend treatments.